

SAM Automation - Literature Review

Context and History of the Problem

Industrial maintenance has undergone a clear evolution over the past decades. **Reactive maintenance (RM)**, or run-to-failure, was the earliest widespread strategy: equipment was repaired only after breakdown, maximizing short-term utilization but resulting in high unplanned downtime, collateral damage, and large repair costs (Zhu et al., 2019). **Preventive maintenance (PM)** emerged to address these risks by scheduling interventions at fixed intervals, based on elapsed time or usage cycles (Zhu et al., 2019). While PM reduced catastrophic failures compared to RM, it introduced inefficiencies through unnecessary part replacements and scheduled downtime. Studies by Zhu et al. (2019) and Murtaza et al. (2024) show that **reactive repairs are typically three times more expensive than preventive actions**, yet preventive strategies still lack precision because they are not tied to the actual condition of equipment. **Predictive maintenance (PdM)** was developed as a response, aiming to perform interventions only when sensor data indicate degradation, balancing repair and prevention costs while maintaining system reliability (Zhu et al., 2019).

The economic importance of this transition is evident. Downtime is highly costly: Amazon reported a loss of **USD 4 million during a 49-minute outage in 2013**, while industry averages estimate losses of **USD 138,000 per hour of downtime** (Zhu et al., 2019). In heavy industry, maintenance expenditures dominate operating costs: operations and maintenance represent **20–35% of revenues** in offshore wind power and **15–70% of production costs** in oil and gas (Zhu et al., 2019). At a macro scale, the International Energy Agency estimates that unplanned outages in energy-intensive sectors cost nearly **USD 50 billion annually**, with facilities losing **3–5% of production capacity** (García et al., 2025).

The technological foundations of PdM emerged from advances in **condition monitoring techniques** such as vibration analysis, thermography, and acoustic monitoring, which enabled the first systematic approaches to fault detection (Zhu et al., 2019). With the onset of **Industry 4.0**, these methods were augmented by the integration of **IoT sensors, big data pipelines, and artificial intelligence algorithms** (Murtaza et al., 2024). IoT devices made it possible to collect continuous, high-frequency operational data, while scalable data infrastructures and analytics transformed raw sensor streams into health indicators and failure predictions (Givnan et al., 2021). This marked the transition of PdM from **isolated condition-monitoring tools to scalable, factory-wide maintenance strategies** (Zhu et al., 2019).

The impact of these developments has been documented across industries. Predictive approaches have been shown to **reduce downtime by up to 50%** and **lower maintenance costs by 10–40%** compared to preventive maintenance (Zhu et al., 2019). A systematic review of implementations from 2017 to 2024 further confirmed that predictive strategies typically cut **downtime by 10–20%** and deliver **global savings of USD 8–15 billion per year** in energy-intensive industries (García et al.,

2025). Beyond direct cost reductions, PdM also decreases spare parts inventory, avoids unnecessary interventions, and increases overall equipment availability and lifetime (Murtaza et al., 2024).

Looking forward, the transition to **Industry 5.0** reframes the role of PdM. Whereas Industry 4.0 positioned predictive maintenance primarily as a technical solution for reducing downtime and costs, Industry 5.0 emphasizes **sustainability, resilience, and human-centric design** (Murtaza et al., 2024). This means PdM is now expected not only to optimize schedules and extend equipment life but also to reduce wasted energy from degraded machinery, contribute to resource efficiency, and provide outputs that are interpretable and usable by technicians in **complex, human-machine collaborative environments** (García et al., 2025).

Review of Existing Solutions, Competitors, and Best Practices

Predictive maintenance solutions are often built on standardized frameworks. **ISO 13374** defines the **Open System Architecture for Condition-Based Monitoring (OSA-CBM)**, which specifies six functional blocks: data acquisition, data manipulation, state detection, health assessment, prognostics, and advisory generation (Zhu et al., 2019). PdM 4.0 architectures expand this model by adding preprocessing, decision support, and maintenance execution modules, integrating IoT devices, cyber-physical systems, and cloud computing for high-frequency, factory-wide deployments (Murtaza et al., 2024; García et al., 2025).

Signal processing remains central to PdM, with **Fast Fourier Transform (FFT)**, **wavelets**, and **CEEMDAN** widely used to extract features from noisy time-series data. These techniques are combined with machine learning models such as **SVMs, CNNs, and LSTMs** for **fault detection** and **Remaining Useful Life (RUL) estimation**. Hybrid models that merge physics-based and data-driven approaches improve robustness under non-stationary conditions and increase interpretability (García et al., 2025).

In robotics, Morettini (2021) demonstrated PdM using only internal sensors. Torque profiles were converted into a joint health index, and anomalies were predicted with **Gaussian Process Regression, Exponentron, and hybrid models**, with hybrid methods performing best. The study also noted reduced accuracy with low sampling rates or missing data. Eang and Lee (2025) proposed a **CNN-RNN hybrid observer** for DC motor drives, analyzing temperature and rotational speed to detect gear tooth wear and misalignments. Their model outperformed **CNN-LSTM** in both accuracy and computational efficiency, supporting real-time deployment. Wojtulewicz and Chaber (2025) integrated PdM directly into robot controllers via **IIoT**. Process data from CNC and robot controllers were analyzed in **PLCs** to calculate component consumption indicators, visualized on HMI panels, and shared via MQTT, achieving predictive diagnostics without additional sensors.

Visualization and interpretability are also advancing. Paraschos et al. (2021) developed **VisioRed**, a tool that combines **PCA-based dimensionality reduction with explanation techniques (LIME)**,

LioNets) to show feature importance in time-series PdM models. It also includes a prescriptive module (XYZ) that recommends sensor value changes to extend RUL.

Real-time anomaly detection has been advanced by approaches using deep learning, in particular **stacked autoencoders (SAEs)**. Givnan et al. (2021) proposed a real-time PdM framework where SAEs are trained on **healthy operational data** to reconstruct input signals; anomalies are detected as deviations measured by **reconstruction error**. Their system introduced a **flexible traffic-light scale** (green for normal, amber for concerning, red for critical) that allows early intervention before breakdown. The framework was designed for edge deployment to **minimize bandwidth and storage** by transmitting only **anomaly-related windows**, making it practical for industrial use cases with high-frequency sensor data.

Industrial competitors such as **ABB, FANUC, and KUKA** embed PdM features in **proprietary platforms** (e.g., ABB Ability, FANUC Zero Downtime, KUKA Connect). Bogue (2025) notes that while these solutions deliver value within their ecosystems, they are **brand-specific**. Independent providers, such as SAM Automation, have an opportunity to differentiate by building vendor-agnostic platforms that unify heterogeneous robot data while adopting best practices from academic and industrial advances.

Gaps in Current Knowledge or Approaches

Despite significant progress, several limitations remain in predictive maintenance research and practice. A major challenge is **data scarcity and imbalance**: industrial datasets contain abundant normal operation data but few fault events, forcing reliance on simulations or lab experiments (Zhu et al., 2019; García et al., 2025). Models must also contend with non-stationary environments, where varying loads, speeds, and noise complicate feature extraction and reduce reliability (García et al., 2025).

Another gap lies in **interpretability**. Deep learning models achieve high accuracy but are often black boxes; tools such as VisioRed provide transparency but are not yet widely adopted in practice (Paraschos et al., 2021). **Integration** challenges persist as well: most industrial PdM platforms (e.g., ABB, FANUC, KUKA) are proprietary and vendor-specific, limiting their applicability in multi-brand environments (Bogue, 2025).

Finally, there is a **deployment gap** between laboratory studies and industrial reality. Many solutions are validated on **controlled datasets** (e.g., NASA turbofan, simulated torque wear) but perform less reliably with missing data, low sampling rates, or fluctuating operating conditions in factories (Morettini, 2021; Givnan et al., 2021). In the shift toward Industry 5.0, PdM must also address sustainability and human-centricity, reducing energy waste and delivering interpretable outputs that technicians can act upon (Murtaza et al., 2024).

To help address these gaps, this project aims to design a **cross-brand, multi-machine predictive maintenance platform**. By combining anomaly detection techniques with robust data pipelines, the system will seek to capture subtle patterns and nuances in robot behavior across heterogeneous environments, moving toward a vendor-agnostic and more resilient PdM solution.

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